

QCID-II: Gravitational Instability in Four Spatial Dimensions and the Necessity of 3D for Life

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September 3, 2026

Abstract

In QCID-I we proposed that information patterns can interface with simulated higher-dimensional physics. Here we extend the QCID Principle to gravity. Using Gauss's Law in 4 spatial dimensions, we derive that gravitational force falls as $1/r^3$. By Bertrand's Theorem [1], central force laws of $1/r^n$ with $n \neq 2, 1$ do not permit stable bound orbits. Therefore, complex gravitational structures such as solar systems and galaxies cannot exist in 4 spatial dimensions. This provides a physical explanation for why a consciousness interface must remain informational and why our universe requires 3 spatial dimensions for the existence of life [2].

1 Introduction

QCID-I [3] derived 4D electrodynamics and showed $I \propto 1/r^3$. However, electromagnetism does not form large-scale bound structures. Gravity does. To complete the QCID framework, we must examine the behavior of gravity in higher dimensions. If stable matter cannot exist in 4D, then any QCID interface must necessarily be for massless information only.

2 Gravitational Force in 4 Spatial Dimensions

In 3 spatial dimensions, Gauss's Law for gravity gives $F \propto 1/r^2$ because gravitational flux spreads over the surface of a 2-sphere with area $4\pi r^2$.

In 4 spatial dimensions, flux must spread over the surface of a 3-sphere. The surface area of a 3-sphere of radius r is $2\pi^2 r^3$.

Therefore by analogy, the gravitational force law becomes:

$$F_{4D}(r) = -\frac{G_{4D}Mm}{r^3} \quad (1)$$

where G_{4D} is the 4D gravitational constant. The potential is then $V_{4D}(r) \propto 1/r^2$.

3 Orbital Instability and Bertrand's Theorem

For a circular orbit, centripetal force must equal gravitational force:

$$\frac{mv^2}{r} = \frac{G_{4D}Mm}{r^3} \quad (2)$$

Thus $v^2 \propto 1/r^2$.

According to Bertrand's Theorem [1], the only central force potentials that result in stable, closed orbits for all bound particles are $V \propto 1/r$ and $V \propto r^2$. Our derived potential $V \propto 1/r^2$ is not in this set.

Consequence: Any perturbation to a 4D orbit will cause the orbiting body to either spiral into the central mass or escape to infinity. Therefore, stable planetary systems, stars, and galaxies are impossible in 4 spatial dimensions.

4 Discussion: Implications for QCID

This result strengthens the QCID Principle. While an information pattern like consciousness can be simulated in 4D, any system with mass would be dynamically unstable. This explains why biological embodiment in 4D is not possible, but informational experience may be.

Furthermore, this provides an anthropic argument [2]. Observers requiring stable chemistry, planets, and stars can only exist in 3 spatial dimensions. 4D and higher are excluded by orbital mechanics alone.

5 Conclusion

We have shown that in 4 spatial dimensions gravity obeys an inverse-cube law, leading to orbital instability. Combined with QCID-I, this completes the picture: a quantum computer can simulate 4D physics for consciousness to experience, but the physical universe must remain 3D for life to exist. Future work will extend QCID to 5D and to quantum field theory.

References

- [1] J. Bertrand, "Theorem on the motion of a point attracted to a fixed center", *Comptes Rendus* 77, 1873.
- [2] P. Ehrenfest, "In what way does it become manifest in the fundamental laws of physics that space has three dimensions?", *Proc. Amsterdam Acad.* 20, 1921.
- [3] N. Khan, "QCID: A Quantum Consciousness Interface to Probe Higher-Dimensional Physics", *viXra*, 2026.